



MAT1320 LINEAR ALGEBRA EXERCISES IX-X

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1. (A points) Let $\vec{u}$ and $\vec{v}$ be two unit vectors. If $\vec{u} + 2\vec{v}$ is orthogonal to $5\vec{u} - 4\vec{v}$, then which of the followings is the angle between the vectors $\vec{u}$ and $\vec{v}$?

a) 60° b) 90° c) 30° d) $\arccos\left(\frac{1}{3}\right)$ e) $\arccos\left(\frac{2}{7}\right)$

$$(\vec{u} + 2\vec{v}) \cdot (5\vec{u} - 4\vec{v}) = 0$$

$$\Rightarrow 5\vec{u} \cdot \vec{u} - 4\vec{u} \cdot \vec{v} + 10\vec{v} \cdot \vec{u} - 8\vec{v} \cdot \vec{v} = 0$$

$$\text{Since } \vec{u} \cdot \vec{u} = |\vec{u}|^2 \text{ and } \vec{v} \cdot \vec{v} = |\vec{v}|^2, \text{ and } |\vec{u}| = |\vec{v}| = 1,$$

$$5 - 4\vec{u} \cdot \vec{v} - 8 = 0 \Rightarrow \vec{u} \cdot \vec{v} = \frac{1}{2}$$

$$\Rightarrow \vec{u} \cdot \vec{v} = |\vec{u}| |\vec{v}| \cos \alpha = \frac{1}{2} \Rightarrow \cos \alpha = \frac{1}{2} \Rightarrow \alpha = 60^\circ$$

$$\begin{vmatrix} 4 & -3 & 7 \\ 1 & 1 & 1 \end{vmatrix}$$

$$(-3-7)i - (4-7)j + (7)k$$

$$\text{Let } \vec{u} = (u_1, u_2, u_3)$$

$$\vec{u} \cdot \vec{u} = 0 \Rightarrow 2u_1 + 0u_2 + 1u_3 = 0 \Rightarrow 2u_1 + u_3 = 0$$

$$\vec{u} \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 4 & -3 & 7 \\ 1 & 1 & 1 \end{vmatrix} = (10, -3, 7)$$

$$\vec{u} \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = (10, -3, 7)$$

$$(b-c)i - (a-c)j + (a-b)k$$

$$u_1 - u_3 = -3 \text{ and } 2u_1 - u_3 \Rightarrow u_1 - 2u_1 = -3 \Rightarrow u_1 = -1$$

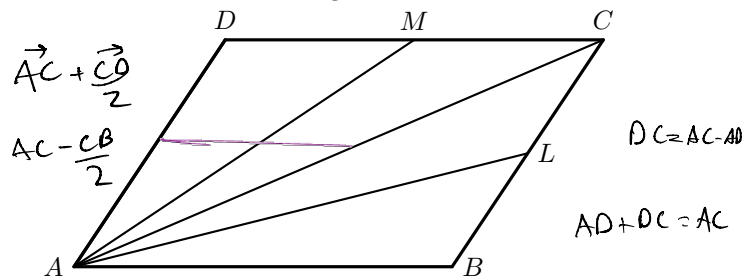
$$u_1 - u_2 = 7 \Rightarrow u_2 = -8$$

2. (D points) Let $\vec{a} = 2\vec{i} + \vec{k}$, $\vec{b} = \vec{i} + \vec{j} + \vec{k}$ and $\vec{c} = 4\vec{i} - 3\vec{j} + 7\vec{k}$. If $\vec{u} \times \vec{b} = \vec{c} \times \vec{b}$ and $\vec{u} \cdot \vec{a} = 0$, then which of the followings is the vector $\vec{u}$?

a) $\vec{i} + 8\vec{j} + \vec{k}$ b) $\vec{i} + 8\vec{j} + 2\vec{k}$ c) $2\vec{i} + \vec{j} - 8\vec{k}$

d) $-\vec{i} - 8\vec{j} + 2\vec{k}$ e) $\vec{i} - 8\vec{j} + \vec{k}$

3. (C points) For the following parallelogram $ABCD$, the points L and M are the middle points of the sides BC and CD , respectively. Then, which of the followings is the vector $\vec{AL} + \vec{AM}$?



a) $\frac{1}{2}AC$ b) AC c) $\frac{3}{2}AC$ d) $2AC$
e) None of them

$$\vec{AL} = \vec{AB} + \vec{BL}$$

$$\vec{AM} = \vec{AD} + \vec{DM}$$

$$\vec{AL} + \vec{AM} = \vec{AB} + \vec{AD} + \vec{BL} + \vec{DM} = \vec{AC} + \frac{1}{2}\vec{AC} = \frac{3}{2}\vec{AC}$$

$$T: 3t^2 \in T, 3t \in T \text{ but } 3t^2 + 3t \notin T$$

$$\Rightarrow T \text{ is not subspace of } P_2$$

$$A: a_1t^2 + b_1t + c_1, a_2t^2 + b_2t + c_2 \in A. \text{ Then,}$$

$$b_1 = 3c_1, b_2 = 3c_2$$

$$\Rightarrow (a_1t^2 + b_1t + c_1) + (a_2t^2 + b_2t + c_2) = (a_1 + a_2)t^2 + (b_1 + b_2)t + c_1 + c_2 \in A$$

$$\forall k \in \mathbb{R},$$

$$\Rightarrow k(a_1t^2 + b_1t + c_1) = ka_1t^2 + kb_1t + kc_1 \in A \text{ since } kb_1 = 3kc_1$$

4. (A points) Let P_2 be the set of all polynomials over real numbers whose degrees are at most 2. Recall that P_2 is a vector space with usual addition and multiplication by a scalar on polynomials. Then, which of the following subsets is a subspace of P_2 ?

$$\mathcal{M} = \{at^2 + bt + c \mid c = 0\}$$

$$\mathcal{A} = \{at^2 + bt + c \mid b = 3c\}$$

$$\mathcal{T} = \{at^2 + bt + c \mid a + b + c = 3\}$$

a) $\mathcal{M}$ and $\mathcal{A}$

b) $\mathcal{M}$ and $\mathcal{T}$

c) $\mathcal{A}$ and $\mathcal{T}$

d) Only $\mathcal{M}$

e) All of them

5. Which of the following subsets are subspaces of the given vector spaces?

$$\mathcal{Y} = \left\{ \begin{bmatrix} x \\ x^2 \end{bmatrix} \mid x \in \mathbb{R} \right\} \subset \mathbb{R}^2$$

$$\begin{matrix} a & b \\ a^2 & b^2 \end{matrix}$$

$$\mathcal{T} = \left\{ \begin{bmatrix} x \\ x+1 \\ 0 \end{bmatrix} \mid x \in \mathbb{R} \right\} \subset \mathbb{R}^3$$

$$\begin{matrix} r, s & \\ a+b & r, s^2 \\ a^2+b^2 & \end{matrix}$$

$$\mathcal{U} = \left\{ \begin{bmatrix} x \\ 0 \\ 0 \end{bmatrix} \mid x \in \mathbb{R} \right\} \subset \mathbb{R}^3$$

a) ~~Only $\mathcal{Y}$~~

b) ~~Only $\mathcal{T}$~~

c) Only $\mathcal{U}$

d) ~~$\mathcal{Y}$ and $\mathcal{T}$~~

e) ~~$\mathcal{T}$ and $\mathcal{U}$~~

$$\mathcal{Y}: \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 4 \end{pmatrix} \in \mathcal{Y} \text{ but } \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \end{pmatrix} \notin \mathcal{Y} \text{ because } 3^2 \neq 5.$$

$$\mathcal{T}: \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix} \in \mathcal{T} \text{ but } \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \\ 0 \end{pmatrix} \notin \mathcal{T} \text{ since } 5 \neq 3+1.$$

$\Rightarrow$ Then, see that $\mathcal{U}$ is closed under addition and multiplication by scalars.
 $\Rightarrow \mathcal{U}$ is a subspace